Chapter 25: Sums of Monotone Operators
Infimal Postcomposition and Composition Operations

Convex Analysis and Monotone Operator Theory

Based on Bauschke & Combettes (2017)

Table of Contents

Overview of Chapter 25

This chapter explores the properties of sums of monotone operators and their compositions:

- **Maximal Monotonicity of Sums**: When is $A + B$ maximally monotone?
- **Local Properties**: Local maximal monotonicity versus global properties
- **3* Monotone Operators**: Special class of monotone operators (Brézis-Haraux)
- **Sum Theorems**: Conditions ensuring sum is maximally monotone
- **Parallel Operations**: Parallel sum and parallel composition
- **Infimal Postcomposition**: Extension of infimal convolution to operators

Key Result: Under appropriate conditions, the sum $A + B$ of two monotone operators is maximally monotone.

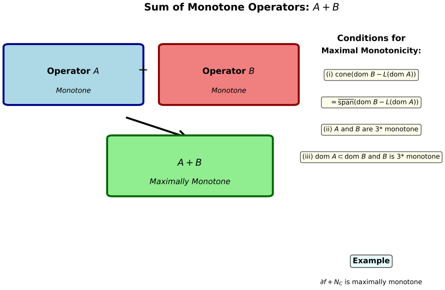
Motivation: Why Study Sums?

Problem Decomposition

- Many optimization problems involve sums: $\min_x f(x) + g(x)$
- Corresponds to monotone operators: $\partial f + \partial g$
- Want to solve using splitting methods

Applications

- Proximal algorithms
- Operator splitting
- Composite optimization
- Duality theory



Canonical Sum and Elementary Properties

Proposition 25.1 (Canonical Sum)
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators. For every $x \in \mathcal{H}$:

$$A(x) + B(x) = \{u + v : u \in A(x),\ v \in B(x)\}$$

Proposition 25.2 (Composite Operators)
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone. Then:

$$\begin{aligned} \mathrm{dom}(A + B) &= \mathrm{dom}A \cap \mathrm{dom}B \\ \mathrm{ran}(A + B) &= \mathrm{ran}A + \mathrm{ran}B \\ \mathrm{gra}(A + B) &= \mathrm{gra}A + \mathrm{gra}B \end{aligned}$$

Sufficient Conditions for Maximal Monotonicity

Theorem 25.3 (Sufficient Condition via Cone)
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ and $B : \mathcal{K} \to 2^{\mathcal{K}}$ be maximally monotone, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, and suppose:

$$\operatorname{cone}(\operatorname{dom} B - L(\operatorname{dom} A)) = \overline{\operatorname{span}}(\operatorname{dom} B - L(\operatorname{dom} A))$$

Then $A + L^* BL$ is maximally monotone.

Example 25.3

$$\text{If } L \in \mathcal{B}(\mathcal{H}, \mathcal{K}),\ A : \mathcal{H} \to 2^{\mathcal{H}},\ B = N_{\operatorname{gra} L}$$

then $A + L^* BL$ is maximally monotone.

Key Sufficient Conditions

Corollary 25.5
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be maximally monotone. If any one of:

(i) $\operatorname{dom} B = \mathcal{H}$

(ii) $\operatorname{dom} A \cap \operatorname{int} \operatorname{dom} B \neq \emptyset$

(iii) $0 \in \operatorname{int}(\operatorname{dom} A - \operatorname{dom} B)$

(iv) $(\forall x \in \mathcal{H})(\exists \varepsilon \in \mathbb{R}_{++})\, [0, \varepsilon x] \subset \operatorname{dom} A - \operatorname{dom} B$

(v) $\operatorname{dom} A$ and $\operatorname{dom} B$ convex, $0 \in \operatorname{sri}(\operatorname{dom} A - \operatorname{dom} B)$

Then $A + B$ is maximally monotone.

Definition 25.7 (Locally Maximally Monotone)
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone and U an open convex subset of $\mathcal{H}$ such that $U \cap \operatorname{ran} A \neq \emptyset$.
A is **locally maximally monotone with respect to** U if: for every $(x, u) \in (\mathcal{H} \times U) \setminus \operatorname{gra} A$, there exists $(y, v) \in (\mathcal{H} \times U) \cap \operatorname{gra} A$ such that

$$\langle x - y \mid u - v \rangle < 0$$

Definition (Globally Locally Maximally Monotone)
A is **locally maximally monotone** if it is locally maximally monotone with respect to every open convex subset of $\mathcal{H}$ with nonempty intersection with $\operatorname{ran} A$.

Relationship to Maximal Monotonicity

Proposition 25.8
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone. Then:

(i) A is locally maximally monotone with respect to $\mathcal{H}$ if and only if A is maximally monotone.

(ii) A is locally maximally monotone if and only if, for every bounded closed convex subset C of $\mathcal{H}$ such that $\operatorname{ran} A \cap \operatorname{int} C \neq \emptyset$ and for every $(x, u) \in (\mathcal{H} \times \operatorname{int} C) \setminus \operatorname{gra} A$, there exists $(y, v) \in (\mathcal{H} \times C) \cap \operatorname{gra} A$ such that $\langle x - y \mid u - v \rangle < 0$.

Proposition 25.9
If $A : \mathcal{H} \to 2^{\mathcal{H}}$ is maximally monotone, then A is locally maximally monotone.

Definition of 3* Monotone Operators

Definition 25.10 (3* Monotone)
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone. Then A is **3* monotone** if:

$$\text{dom } A \times \text{ran } A \subset \text{dom } F_A$$

where $F_A(x, y) = \sup_{(z,w)\in \text{gra } A}(\langle x - z \mid w\rangle + \langle z - y \mid w\rangle)$.

Interpretation: The graph of A has a special rectangular structure in the product space.

Characterizations of 3* Monotone Operators

Proposition 25.11
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be maximally monotone and 3* monotone. Then:

$$\text{dom } F_A = \text{dom } A \times \overline{\text{ran}} A$$

Proposition 25.12
If A is 3-cyclically monotone, then A is 3* monotone.

Examples 25.13 & 25.14

- If $f : \mathcal{H} \to [-\infty, +\infty]$ is proper, then ∂f is 3* monotone.
- If C is a nonempty closed convex subset of $\mathcal{H}$, then N_C is 3* monotone.

Proposition 25.16

Let $A \in \mathcal{B}(\mathcal{H})$ be monotone. The following are equivalent for some $\beta \in \mathbb{R}_{++}$:

(i) A is 3* monotone.

(ii) A is β-cocoercive.

(iii) A^* is β-cocoercive.

(iv) A^* is 3* monotone.

Python Example: Detecting 3* Monotone Operators

```
# For a matrix A (operator in finite dimensions)
# Check if A is beta-cocoercive (hence 3* monotone)
import numpy as np

def is_cocoercive(A, beta):
    """Check if operator A is beta-cocoercive"""
    eigenvalues = np.linalg.eigvals(A + A.T) / 2
    min_eigenval = np.min(eigenvalues)
    return min_eigenval >= beta
```

Theorem 25.23 (Simons)
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators such that $A + B$ is maximally monotone. Suppose that:

$$(\forall u \in \text{ran } A)(\forall v \in \text{ran } B)(\exists x \in \mathcal{H}) \quad (x, u) \in \text{dom } F_A \text{ and } (x, v) \in \text{dom } F_B$$

Then:

$$\overline{\text{ran } A + \text{ran } B} = \overline{\text{ran}(A + B)} \quad \text{and} \quad \text{intran}(A + B) = \text{int}(\text{ran } A + \text{ran } B)$$

This characterizes the range of the sum in terms of the sum of ranges.

Theorem 25.24 (Brézis-Haraux)
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators such that $A + B$ is maximally monotone.
If one of the following conditions holds:

(i) A and B are 3* monotone.

(ii) $\operatorname{dom} A \subset \operatorname{dom} B$ and B is 3* monotone.

Then:

$$\overline{\operatorname{ran}(A+B)} = \overline{\operatorname{ran} A + \operatorname{ran} B} \quad \text{and} \quad \operatorname{int} \operatorname{ran}(A+B) = \operatorname{int}(\operatorname{ran} A + \operatorname{ran} B)$$

Applications of Brézis-Haraux

Corollary 25.27
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators such that $A + B$ is maximally monotone, A or B is surjective, and one of:

(i) A and B are 3* monotone.

(ii) dom $A \subset$ dom B and B is 3* monotone.

Then $A + B$ is surjective (i.e., $A + B$ is in one-to-one correspondence).

Example 25.25: If C and D are closed convex cones, then $N_C + N_D = \mathbb{R} \times \{0\}$ fails unless both are maximally monotone, showing necessity of assumptions.

Parallel Sum Operation

Definition 25.29 (Parallel Sum)
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be operators. The **parallel sum** of A and B is:

$$A \Box B = (A^{-1} + B^{-1})^{-1}$$

Motivation: In network/circuit theory, parallel combination of impedances/conductances.

Connection to Infimal Convolution: For functions $f, g : \mathcal{H} \to [-\infty, +\infty]$:

$$(f \Box g) = (\partial f) \Box (\partial g)$$

Properties of Parallel Sum

Proposition 25.30

Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be operators, and let $x, u \in \mathcal{H}$. Then:

(i) $(A \square B)x = \bigcup_{y \in \mathcal{H}} (A(y) \cap B(x - y))$

(ii) $u \in (A \square B)x \Leftrightarrow (\exists y \in \mathcal{H})\, y \in A^{-1}u$ and $x - y \in B^{-1}u$

(iii) $\operatorname{dom}(A \square B) = \operatorname{ran}(A^{-1} + B^{-1})$

(iv) $\operatorname{ran}(A \square B) = \operatorname{ran} A \cap \operatorname{ran} B$

(v) If A, B are monotone, then $A \square B$ is monotone.

Parallel Sum for Convex Functions

Proposition 25.32
Let $f, g \in \Gamma_0(\mathcal{H})$ and suppose $0 \in \operatorname{sri}(\operatorname{dom} f^* - \operatorname{dom} g^*)$. Then:

$$(\partial f) \square (\partial g) = \partial(f \square g)$$

where the parallel sum of functions is:

$$(f \square g)(x) := \inf_{y \in \mathcal{H}} f(y) + g(x - y)$$

Properties:

- $f \square g \in \Gamma_0(\mathcal{H})$
- $\operatorname{dom}(f \square g) = \operatorname{dom} f + \operatorname{dom} g$

Numerical Example: Parallel Sum

Example: Quadratic functions and their parallel sum

```
import numpy as np

# Define quadratic functions f(x) = (1/2)||x||^2
# and g(x) = (1/2)||x - c||^2

def parallel_sum_quadratic(x, c, lambda_f=1.0, lambda_g=1.0):
    """Parallel sum of two quadratic functions"""
    # For quadratics: (f square g)(x) is also quadratic
    # Parameters combine harmonically
    lambda_sum = 1 / (1/lambda_f + 1/lambda_g)
    center = lambda_sum * c / lambda_g
    return 0.5 * lambda_sum * (np.linalg.norm(x - center) ** 2)

# Example with c = [1, 0]
x = np.array([0.5, 0.5])
c = np.array([1.0, 0.0])
result = parallel_sum_quadratic(x, c)
print(f"Parallel sum at x: {result}")
```

Parallel Composition Definition

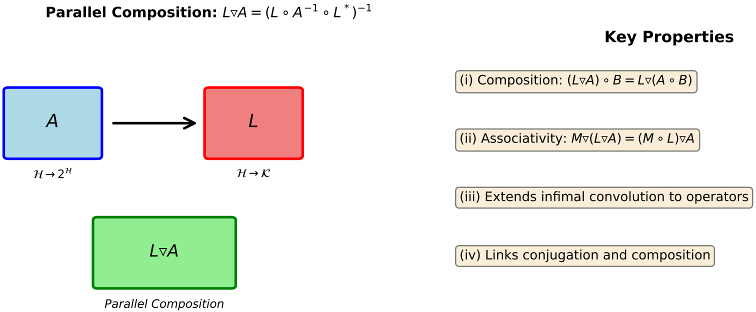
Definition 25.39 (Parallel Composition)

Let $\mathcal{K}$ be a real Hilbert space, $A : \mathcal{H} \to 2^{\mathcal{H}}$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. The **parallel composition** of A by L is the operator from $\mathcal{K}$ to $2^{\mathcal{K}}$:

$$L \triangledown A = (L \circ A^{-1} \circ L^*)^{-1}$$

Alternative names:

- Infimal postcomposition
- Pushforward with infimal convolution



Parallel Composition Properties

Proposition 25.41

Let $\mathcal{K}$ be a real Hilbert space, $A : \mathcal{H} \to 2^{\mathcal{H}}$, $B : \mathcal{K} \to 2^{\mathcal{K}}$ be monotone, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. Then:

(i) $(L \triangledown A) \square B)^{-1} = L \circ A^{-1} \circ L^* + B^{-1}$

(ii) If A, B monotone, then $(L \triangledown A) \square B$ is monotone.

(iii) If A, B maximally monotone and $\text{cone}(\text{ran } A - L^*(\text{ran } B)) = \text{span}(\text{ran } A - L^*(\text{ran } B))$, then $(L \triangledown A) \square B$ is maximally monotone.

Proposition 25.42

Let $G, \mathcal{K}$ be real Hilbert spaces, $A : \mathcal{H} \to 2^{\mathcal{H}}$, $B : \mathcal{H} \to 2^{\mathcal{H}}$, $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$, $M \in \mathcal{B}(\mathcal{G}, \mathcal{K})$. Then:

(i) $L \triangledown (A \Box B) = (L \triangledown A) \Box (L \triangledown B)$

(ii) $M \triangledown (L \triangledown A) = (M \circ L) \triangledown A$

These are the main composition properties for parallel composition of operators.

Infimal Postcomposition for Convex Functions

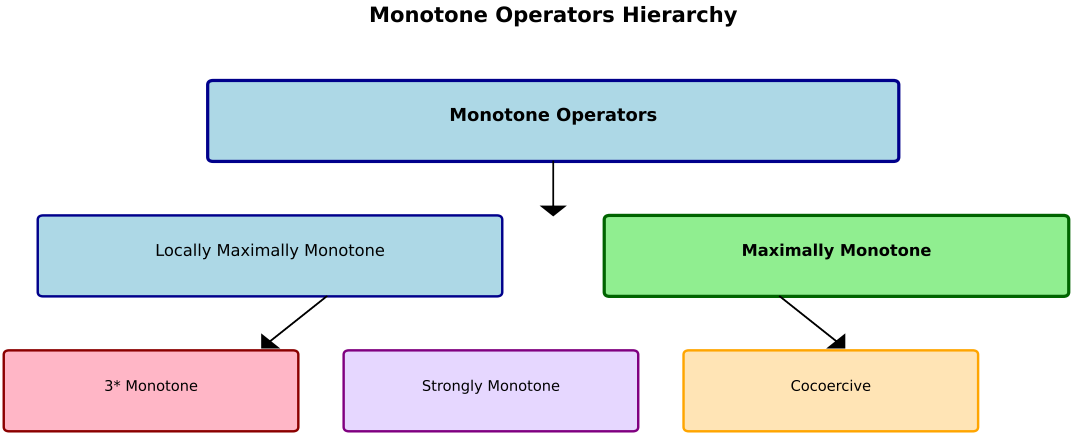
Proposition 25.43
Let $\mathcal{K}$ be a real Hilbert space, $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that $0 \in \operatorname{sri}(\operatorname{dom} f^* - L^*(\operatorname{dom} g^*))$. Then:

(i) $L \triangledown f \in \Gamma_0(\mathcal{K})$

(ii) $\partial(L \triangledown f) \square g = (L \triangledown \partial f) \square g$

This links infimal postcomposition for functions to that for operators.

Operator Hierarchy and Examples



Important Examples

Key Examples of 3* Monotone Operators

Example 25.15: Conditions for 3* Monotonicity
Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone. Then $\operatorname{dom} A \times \mathcal{H} \subset \operatorname{dom} F_A$ and A is 3* monotone in each case:

(i) $(\forall x \in \operatorname{dom} A)\ \lim_{\rho \to +\infty} \inf_{\substack{(z,w) \in \operatorname{gra} A \\ \|z\| \geq \rho}} \frac{\langle z - x \mid w \rangle}{\|z\|} = +\infty$

(ii) $\operatorname{dom} A$ is bounded.

(iii) A is uniformly monotone with a supercoercive modulus.

(iv) A is strongly monotone.

Example 25.17: LL^* is 3* monotone.

Composite Operators: $(T \circ A)$ and $(A \circ T)$

For $T : \mathcal{H} \to \mathcal{H}$ nonexpansive and $A : \mathcal{H} \to 2^{\mathcal{H}}$ monotone:

- **Precomposition**: $T \circ A$ (operator before A)
- **Postcomposition**: $A \circ T$ (operator after T)

Example 25.18: If R is the cyclic right-shift operator, then $\mathrm{Id} - R$ is 3* monotone.

Example 25.20: Let D be nonempty, $T : D \to \mathcal{H}$, $A : \mathcal{H} \to 2^{\mathcal{H}}$ monotone, $\beta, \gamma \in \mathbb{R}_{++}$.
If any one of (i)-(v) holds, then T is 3* monotone:

(i) T is β-cocoercive.

(ii) T is firmly nonexpansive.

(iii) $T = J_{\gamma A}$ (resolvent).

(iv) $T = \gamma A$ (scaled operator).

(v) $\mathrm{Id} - T$ is nonexpansive.

Proposition 25.19: Characterization of 3* Monotone Operators

Proposition 25.19

Let $A : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone and let $\gamma \in \mathbb{R}_{++}$. Then:

(i) A is 3* monotone if and only if A^{-1} is 3* monotone.

(ii) A is 3* monotone if and only if γA is 3* monotone.

This shows 3* monotonicity is preserved under inversion and scaling.

Composition Flow: From Functions to Operators

Compositions and Postcomposition Operations

Infimal Conv.
$f \circ g$

→

Postcomposition
$L \triangledown f$

→

Infimal Postcomp.
$L \triangledown A$

Function Level: Convex Functions

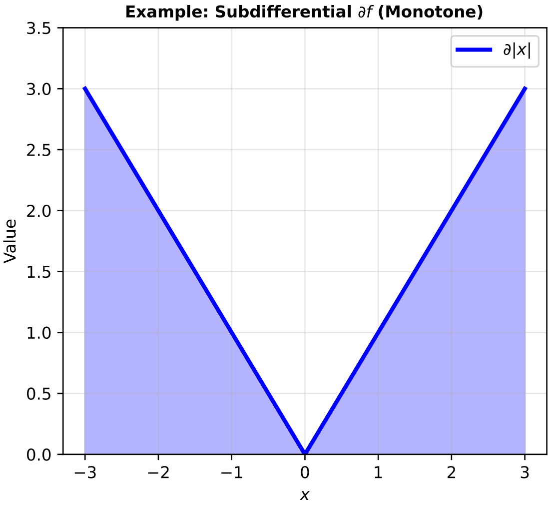
Operator Level: Monotone Operators

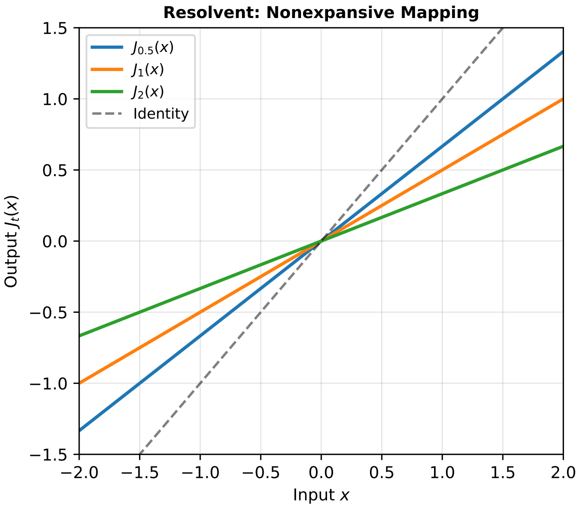
Sum: $A + B$
(infimal conv.)

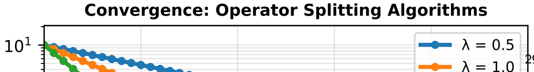
Composition: $L \circ A$
(postcomposition)

Parallel: $L \triangledown A$
(infimal postcomp.)

Numerical Examples







Computing with Monotone Operators: Douglas-Rachford

Problem: Find $x \in \mathcal{H}$ such that $0 \in (A + B)x$ where A, B maximally monotone.

Douglas-Rachford Splitting:

```
def douglas_rachford(A_inv, B_inv, x0, gamma, max_iter=100, tol=1e-6):
    """
    Solve 0 in (A + B)x using Douglas-Rachford splitting.
    Assumes we can compute A^{-1} and B^{-1}.
    """
    x = x0
    for k in range(max_iter):
        # Apply resolvents: J_A = (I + A)^{-1}, J_B = (I + B)^{-1}
        y = A_inv(x)                    # y = J_A(x)
        z = 2 * B_inv(y) - y            # z = 2*J_B(y) - y
        x_new = x + gamma * (z - y)     # x := x + gamma(z - y)

        if np.linalg.norm(x_new - x) < tol:
            break
        x = x_new
    return x
```

Key: Sum of monotone operators allows operator splitting.

Parallel Sum Algorithm

Problem: Compute $(A \square B)x$ for operators A, B given x.

```
def parallel_sum(A_inv, B_inv, x, solver_tol=1e-6):
    """
    Compute u in (A square B) x.
    Based on: u in (A square B)x iff
              exists y in H such that y in A^{-1}u and x-y in B^{-1}u
    """
    # Solve for u such that A_inv(u) + B_inv(u) contains a pair
    # summing to x. This is equivalent to solving:
    # 0 in (A^{-1} + B^{-1})u - x

    def residual(u):
        return A_inv(u) + B_inv(u) - x

    # Use fixed-point iteration or Newton-type method
    u = np.zeros_like(x)  # Initial guess
    for iteration in range(100):
        res = residual(u)
        if np.linalg.norm(res) < solver_tol:
            break
        # Simplified gradient step (in practice use more sophisticated solver)
        u = u - 0.1 * res

    return u
```

Proposition 25.33: Resolvent Relations

Proposition 25.33
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators. Then:

$$J_A \Box J_B = J_{(1/2)(A+B)} \circ \frac{1}{2}\mathrm{Id}$$

where $J_A = (\mathrm{Id} + A)^{-1}$ is the resolvent (Yosida approximation).

Corollary 25.34
Let $A, B : \mathcal{H} \to 2^{\mathcal{H}}$ be monotone operators. Then:

$$J_{A+B} = (J_{2A} \Box J_{2B}) \circ 2\mathrm{Id}$$

Example: Computing Subdifferentials

Python Example: Subdifferential of Maximum Function

```python
import numpy as np

def subdiff_max_convex(x, functions_list):
    """
    Compute subdifferential of max(f_i(x)) using sum decomposition.
    """
    n_funcs = len(functions_list)
    values = np.array([f(x) for f in functions_list])
    max_val = np.max(values)

    # Active functions (those achieving the maximum)
    active_indices = np.where(np.abs(values - max_val) < 1e-6)[0]

    # Subdifferential is convex combination of subdifferentials
    # of active functions
    subdiff = np.zeros_like(x)
    for idx in active_indices:
        # Compute gradient/subgradient of f_idx
        grad_f = compute_gradient(functions_list[idx], x)
        subdiff += grad_f / len(active_indices)

    return subdiff
```

This uses the fact that subdifferentials combine via monotone operator sums.

Key Theorems Summary

1. **Theorem 25.3**: Cone condition ensures $A + L^*BL$ is maximally monotone.
2. **Corollary 25.5**: Practical conditions for maximal monotonicity of $A + B$.
3. **Proposition 25.8**: Relationship between local and global maximal monotonicity.
4. **Proposition 25.19**: 3* monotonicity characterized via cocoercivity.
5. **Theorem 25.24 (Brézis-Haraux)**: Range characterization for sum of 3* operators.
6. **Definition 25.29**: Parallel sum via inverse sum formula.
7. **Definition 25.39**: Parallel composition and infimal postcomposition.
8. **Proposition 25.42**: Composition properties for parallel operators.

Applications and Algorithms

Optimization Splitting Methods

- Douglas-Rachford algorithm for $\min_x f(x) + g(x)$
- Forward-backward splitting
- Proximal alternating direction method of multipliers (PADMM)

Duality and Saddle Points

- Monotone operator framework for duality
- Saddle point problems via sum of operators

Network and Distributed Optimization

- Parallel composition for network operators
- Decentralized algorithms

Machine Learning Applications

- Composite optimization: $\min_x f(x) + g(Ax)$
- Proximal methods for regularization

Summary: Key Insights

- **Sums of Monotone Operators**: Under appropriate conditions (cone condition, 3* monotonicity), the sum $A + B$ preserves maximal monotonicity.
- **3* Monotonicity**: A restrictive but important class including subdifferentials and normal cones, characterized by cocoercivity for linear operators.
- **Brézis-Haraux Theorem**: Provides range characterization when one operator is 3* monotone.
- **Parallel Operations**: Extend composition and convolution to operators.
- **Infimal Postcomposition**: Links function-level concepts (infimal convolution) to operator level, enabling systematic study of composite structures.
- **Practical Impact**: Enables design of splitting algorithms for complex optimization problems.

Exercises I

1. Show that if A and B are monotone and $\operatorname{dom} B = \mathcal{H}$, then $A + B$ is monotone.
2. Let $f : \mathcal{H} \to [-\infty, +\infty]$ be convex. Show that ∂f is 3* monotone.
3. Verify that for the cyclic right-shift operator R, the operator $\mathrm{Id} - R$ is 3* monotone.
4. Compute the parallel sum $(A \Box B)$ for the case where A and B are projections onto closed convex sets.
5. Prove that if $L : \mathcal{H} \to \mathcal{K}$ is a linear operator and $A : \mathcal{H} \to 2^{\mathcal{H}}$ is monotone, then $L \triangledown A : \mathcal{K} \to 2^{\mathcal{K}}$ defined by $(L \triangledown A) = (L \circ A^{-1} \circ L^*)^{-1}$ is monotone.
6. Show that Corollary 25.27 applies to the case of proximal operators combined with projections.
7. Implement Douglas-Rachford algorithm for solving $0 \in (A + B)x$ numerically in Python (see Example 25.34 for resolvent relations).